

Peer review, mapped onto the literature review

A live review of this project's literature review, mapped section by section onto what each criticism actually changed. Quotes are attributed by content.

1. The central critique

"I feel that you are trying to justify your hypothesis with papers. [...] new papers ideally should be giving birth to new hypotheses or experiments."

This is the one criticism that lands on the method rather than on the findings, and it is correct about the **order of operations**:

```
transcript -> 5 clusters -> 11 seeds -> H1-H6 -> 590 candidates -> 58 -> map each to an
existing H

                                ^                               ^
                                hypotheses fixed here           literature only

sorted, never allowed to add
```

`shortlist.csv` has `maps_to_H` and `verdict` columns. Both measure *fit to a hypothesis that already existed*. Nothing in the pipeline can emit "this paper implies a seventh hypothesis" — there is no column for it, so the answer could never appear. The `DROP` token and the cluster quotas compound it: a paper with no home in A–E leaves the funnel.

What this does not invalidate. The corpus, the extraction and the findings are unaffected — 58 papers were selected against clusters (which came from humans playing the game, not from H1–H6), and the nine Elicit columns are descriptive, not confirmatory. The protocol confound, the open-weight count and Rodionov's masking result were all found *against* prior expectation, which is the signature of a review that was not merely confirming itself. The defect is narrower than "the review is biased": it is that **the review had no output channel for a hypothesis it did not start with**.

Fix. One backward pass over the 58, asking the opposite question of each: *what experiment does this paper suggest that H1–H6 does not already name?* New column `suggests_experiment` in `shortlist_extracted.csv`. Cheap — the extraction is already done; this re-reads the mechanism and result columns with the question inverted.

2. "How can you say H3 and H5 are unclaimed?"

Challenged directly, and the challenge is fair as stated. The brief's evidence is **6 papers for H3 and 7 for H5** — the thinnest coverage in the shortlist. Thin coverage in a corpus is not absence in the field. The review already documents the reason to distrust its own coverage here: the citation graph surfaced **none** of the 13 ARC-AGI-3 papers across 512 backward references and 8 forward lists, because no seed cites the benchmark. Both were found only by direct name search.

The honest form of the claim, which is what the milestone brief should say:

No ARC-AGI-3 paper *in this corpus* isolates controllability (H3) or visited-state pruning (H5), and the corpus is known to under-sample this benchmark. Treat as "not found", not "not done".

The mentor's own resolution in the room was the same one: **don't spend the milestone defending unclaimed-ness, spend it running the experiment.** Whether H5 is novel does not change whether the harness improves the score.

3. "H5 and H6 are a single hypothesis"

"I frankly think that your hypothesis five and six are a single hypothesis. They might fall in a different bucket of improvement, but ideally they are one experiment."

Partly right, and the extraction says which part. Both are *learning from your own past trajectory to avoid repeating it*. They differ in what is stored and when it fires:

	H5	H6
stored	visited-state signatures	failure episodes
fires	before acting, to prune	after a reset, as a rule
lifetime	within an episode	across episodes
lineage	Go-Explore	episodic-to-semantic consolidation

As **mechanisms** they are separable. As an **experiment** the mentor is right: one harness carrying an episodic store serves both, and splitting them into two runs doubles the compute for a distinction the metric may not resolve. Run them as one treatment with two ablation switches; report the deltas separately.

This also dissolves the H5-vs-H6 choice the room spent time on. It is not a choice.

4. Hypotheses must become falsifiable experiments

"That experiment should clearly state: this is my hypothesis, my experiment, this hypothesis is falsified if this happens. [...] You remove the harness, your SLM comes in, you test it, you have a baseline. You test it by adding this harness. Does this improve or not? If it does not improve, your experiment gets denied."

This is the missing artefact. H1–H6 are currently *claims* — each has a mechanism and a metric, but no stop condition and no baseline. The brief names metrics (actions-to-solve, repeat-hazard rate) and never a falsifier.

Deliverable: `research_prep/notes/experiment_register.md`, one block per experiment —

E1	hypothesis	H5+H6 as one treatment
	baseline	frozen open-weight SLM, no harness, N seeds
	treatment	same weights, same prompts, harness on
	primary metric	actions-to-solve on held-out levels
	falsified if	treatment does not beat baseline outside seed variance
	abandoned if	>X GPU-hours, or the delta is inside noise after N seeds
	cross-cluster	report delta on every cluster, not only the target

This replaces the ranking write-up as the next deliverable. The ranking was going to argue *which hypothesis is most promising*; the register commits to *what would prove it wrong*, which is what the milestone actually needs.

5. Track the cross-cluster delta, not just the headline

"That second experiment helped one of your clusters but regressed on another [...] we can do that detailed review of that delta — why it happened like that."

A new requirement on the evaluation harness, and it lands on an item open since 2 September: the efficiency-ratio field in `src/evaluation/metrics.py`. Per-run output needs to be a **vector over clusters A–E**, not a scalar. A harness that improves H6 by caching failure rules may cost exploration breadth (cluster C) — invisible under a single RHAE number.

6. Widen the sources past the ML literature

"I will not just look from the technical perspective. I will also start looking at research topics from the fundamental question perspective — how do people start to understand the world? Psychology, cybernetics. [...] How do you combine two different worlds into a single world?"

The sharpest *additive* suggestion, and the review's clearest gap. The five clusters were derived from **humans playing the game** — the peer-discussion transcript — and were then answered exclusively from ML papers reachable by citation graph from ML seeds. The human-solving literature that generated the questions was never consulted.

Concretely unsearched, one per cluster:

Cluster question	Unsearched literature
How does the agent understand an unknown world?	developmental psychology; infant object exploration; Gibsonian affordances
What is probably true?	Bayesian cognitive science; children's causal learning (Gopnik)
How is useful information discovered?	cybernetics — Ashby's requisite variety, Powers' perceptual control
Knowledge into action	motor-learning and skill-acquisition literature
Improving over experience	learning-curve and transfer literature outside RL

The mentor's second technique — **reverse-engineer a landmark paper's thought process** rather than read its citations — is a different sampling method from the citation graph, and would have been immune to the blind spot in §2. The citation graph finds what an idea *led to*; this finds what it *came from*.

Scope caution, accepted in the room: this is a second review, not a patch to this one. The right size for the milestone is a **bounded 2–3 hour pass, ~15–20 sources, one per cluster question**, filed as a `cluster_origins` appendix. It is not another 590-candidate funnel.

7. The model decision — closed

Settled in the room, and it matches what the review found:

- Freeze an open-weight SLM (Qwen or Gemma), **no weight changes**; the harness is the independent variable.
- Inference now from a hosted copy (RunPod); download and upload the weights as a Kaggle dataset for submission.

- Kaggle allows uploaded model weights. Pre-training is allowed; only the submitted run is offline.

The review supports this from an angle nobody in the room raised. **Rodionov found capability gains exceed the differences between architectural variants** — at frontier scale, mechanism comparisons are masked. A frozen small model is therefore not a Kaggle limitation to work around; it is the *condition under which the harness delta is measurable at all*. The mentor arrived at frozen-baseline-vs-harness on engineering instinct; the literature says it is also the only regime where the answer means anything.

This also answers the question left hanging in the room — "I am not sure whether harness will help or fine-tuning will help." Evidence exists: SAGE runs 7 backbones from 1B–7B with a gate that **resolves in closed form without an LLM call**; PREMem ladders 3B–72B. 13 of 56 papers ran open-weight models. None on ARC-AGI-3 — which is the gap, stated precisely.

8. Stop reviewing

"Just maybe one hundred papers or something and come up with a set of experiments. Once you have the experiments, you're almost all sorted."

58 papers, 56 extracted, is inside that. **The review is done**. Everything above that adds reading — the backward pass (§1), the cross-domain pass (§6) — is bounded and time-boxed, or it does not happen. The next artefact is the experiment register, not more literature.

9. Three asks aimed at peers

a. Package the method as a reusable skill.

"If you can have that as a skill file — like Ramadan delivered a Genesis structure that helps anyone start a project — similarly, a true research harness. Given any problem, can somebody come up with a set of hypotheses to try? That is the whole idea."

The pipeline is already scripted (`fetch_references.py` , `rank_candidates.py` , `merge_forward.py` , `join_elicit.py`) and the method note documents it. What is missing is the wrapper that makes it runnable on someone else's problem. A real deliverable, and mostly assembly rather than new work.

b. A plain-language layer on the findings. The mentor pasted the findings artifact into ChatGPT asking it to "assume I'm in class five" and explain each insight against real ARC-AGI-3 examples. That the room needed that step is feedback on the artifact: the findings are correct but land as claims about literature rather than claims about the game. A worked example per insight — *here is the plus sign, here is what H3 would do differently* — fixes it.

c. Story over leaderboard.

"I do not care whether you are top of the leaderboard. I want to see what you have done behind. That's what the story should count."

Explicit validation of the revision-log style already used in `literature_review_plan.md` — the co-citation rule that failed, the blind spot, the corrections. Keep recording failures in place rather than tidying them away.

10. Ideas carried over from a parallel review

Three ideas came up alongside this one. Two transfer; one does not.

- **Genesis for the experiment loop.** The mentor's caveat is the important part: "*Genesis has to be built for the experiments. I have not built it like that.*" Adapt the workflow before using it — spec/gate/approve maps onto hypothesis/run/verdict, but nothing in Genesis knows what a falsifier is.
 - **Small, reviewable units.** A one-task-one-PR discipline is the right shape for experiments: one experiment, one commit, one recorded verdict, no batching.
 - **Repository indexing and agent-readiness** (grep-tile, Factory levels) — a useful thread for the enterprise-agent build, not for this one. Noted and parked.
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Revised pending list

	Item	Change
1	Experiment register with falsifiers	new — replaces the ranking write-up as next deliverable
2	Kaggle account + phone verification	unchanged, still the only external blocker
3	H5+H6 merged into one treatment, two ablation switches	new
4	Cluster-vector metrics in <code>src/evaluation/metrics.py</code>	reframed — was a scalar efficiency ratio
5	Backward pass: <code>suggests_experiment</code> over the 58	new, ~1 session
6	Cross-domain pass, time-boxed, ~15–20 sources	new, bounded
7	Soften the H3/H5 "unclaimed" claim to "not found in corpus"	new — one edit to the brief, one to an artifact
8	Freeze Qwen/Gemma, RunPod inference	closes the open model decision
9	Research-harness skill file	new, peer-facing
10	Worked ARC examples in the findings artifact	new
11	Ranking write-up (<code>literature_review.md</code>)	demoted — the register supersedes it
12	Move the share pin on artifact 54461d0b	unchanged, sits with you